

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Bachelor of Technology (Computer Science and Engineering) SEMESTER - 5 Summer 2025 (Supply.)

Course :Bachelor of Technology (Computer Science and Engineering) Branch : Engineering and Technology

Semester : SEMESTER - 5

Subject Code & Name: BTCOC501 - DATABASE SYSTEMS

Time : 3 Hours]

[Total Marks : 60

Instructions to the Students:

1. Each question carries 12 marks.
2. Question No. 1 will be compulsory and include objective-type questions.
3. Candidates are required to attempt any four questions from Question No. 2 to Question No.6
4. Use of non-programmable scientific calculators is allowed.
5. Assume suitable data wherever necessary and mention it clearly.

Q1. Objective type questions. (Compulsory Question)

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- 1 Which of the following is a characteristic of a good database schema?
 - a) Redundancy
 - b) Data inconsistency
 - c) Atomicity
 - d) Data integrity
- 2 Which of the following is the most general definition of a database?
 - a) A collection of files
 - b) A collection of related data
 - c) A collection of software
 - d) A collection of hardware
- 3 What does the 'PROJECT' operation do in relational algebra?
 - a) Selects rows based on a condition
 - b) Selects columns from a relation
 - c) Renames a relation
 - d) Combines two relations
- 4 What is the purpose of the 'HAVING' clause in SQL?
 - a) To filter rows before grouping
 - b) To filter rows after grouping
 - c) To sort the result set
 - d) To specify the join condition
- 5 Which of the following is NOT a DDL command?
 - a) CREATE
 - b) ALTER
 - c) DELETE
 - d) TRUNCATE
- 6 Which of the following is a characteristic of a relational database?
 - a) Data is stored in a hierarchical structure.
 - b) Data is stored in a network structure
 - c) Data is stored in tables
 - d) Data is stored in a graph structure
- 7 Which of the following can result from concurrent transactions without proper isolation?
 - a) Deadlock
 - b) Dirty read
 - c) Redundancy
 - d) Normalization
- 8 What is the purpose of the log in a DBMS transaction management system?
 - a) To store the database schema
 - b) To record all changes made to the data
 - c) To optimize queries
 - d) To normalize data

- 9 Which schedule is considered correct in terms of concurrency control?
- Non-serial schedule
 - Serializable schedule
 - Conflict-free schedule
 - Lock-free schedule
- 10 Which of the following is not a concurrency control protocol?
- Two-phase locking
 - Time-stamp ordering
 - Validation-based
 - Shadow paging
- Which of the following aggregate functions returns the number of rows in a table?
- SUM
 - AVG
 - COUNT
 - MAX
- 12 Which of the following is used to perform a selection operation in relational algebra?
- Pi
 - Sigma
 - Rho
 - Join

Q2. Solve the following.

- Explain the difference between logical and physical data independence. Provide a real-world example to illustrate the importance of logical data independence in database evolution. 6
- Consider a University database. Design an E-R diagram to represent the entities like Students, Courses, Professors, and Departments. Include relevant attributes and relationships between these entities. 6

Q3. Solve the following.

- Differentiate between Tuple Relational Calculus (TRC) and Domain Relational Calculus (DRC). Give examples of when one might be preferred over the other. 6
- Define and Explain 1. Primary Key 2. Candidate Key 3. Superkey 6

Q4. Solve Any Two of the following.

- The retail company wants to analyze sales data from different stores. The data includes store ID, product ID, sales date, quantity sold, and sales amount. Write an SQL query to perform the following analysis: 1. Find the total sales amount for each store. 2. Identify the top 5 products sold across all stores. 3. Find the stores that have sold more than 100 units of a specific product. 6
- Explain with examples the purpose of outerjoins, semi-joins, and division. Provide examples of scenarios where these operations are particularly useful. 6
- Write relational algebra expressions to: 1. Find the names of all employees who work in the Sales department. 2. Find the average salary of employees in each department. 6

Q5. Solve Any Two of the following.

- Explain Functional Dependency Theory with examples. 6
- Explain the structure and working of a B+ Tree index file. 6

- C) Given a relation and set of FDs, normalize it step-by-step into BCNF or 4NF. 6
R(StudentID, CourseID, Instructor, InstructorPhone, Hobby)
And the following dependencies:
FD1: StudentID $\rightarrow$ Hobby
FD2: CourseID $\rightarrow$ Instructor, InstructorPhone
FD3: Instructor $\rightarrow$ InstructorPhone
MVD1: StudentID $\twoheadrightarrow$ Hobby (each student may have multiple hobbies independently of other attributes)

Q6. Solve Any Two of the following.

- A) Explain ACID properties with suitable examples. 6
B) What is serializability? Explain conflict and view serializability with examples. 6
C) Discuss the transaction processing architecture and how DBMS handles failures. 6

*** End ***